

Homework #7

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Due Date: November 10th, 2016

Problem #1

$$(65536 - x) \times 1.085 \mu\text{s} = 5 \text{ ms} = 5000 \mu\text{s} \quad (1)$$

$$= 60928_{10} \quad (2)$$

$$= EE00_{16} \quad (3)$$

Therefore, TH1 = EE_{16} and TL1 = 00_{16}

Problem #2

Assuming Problem #1 is correct,

```
ORG 0H

MOV R0, #100D
LOOP:  MOV TMOD, #01D
        MOV TH0, #EEH
        MOV TLO, #00H    ; (FFFF - EE00 + 1)*1.085 = 5ms
        ACALL DELAY
        DJNZ R0, LOOP    ; 100*5ms = 500 ms

        AJMP DONE

DELAY:  SETB TR0
LOOP2:  JNB TF0, LOOP2
        CLR TR0
        CLR TF0
        RET

DONE:   NOP
        END
```

Problem #3

5 kHz = 500 μ s. So,

$$TH1 = EE_{16} \quad \text{and} \quad TL1 = 00_{16} \quad (4)$$

The code is as follows,

```
ORG OH

MOV R0, #100D
CLR P0.0

LOOP:  MOV TMOD, #01D
        MOV TH0, #EEH
        MOV TLO, #00H    ; (FFFF - EE00 + 1)*1.085 = 5ms
        ACALL DELAY

        CJNE R0, #75D, SKIP
        SETB P0.0

SKIP:   DJNZ R0, LOOP    ; 100*5ms = 500 ms

        AJMP DONE

DELAY:  SETB TR0
LOOP2:  JNB TF0, LOOP2
        CLR TR0
        CLR TF0
        RET

DONE:   NOP
        END
```

Problem #4

Because we know frequency $(f) = \frac{1}{\text{period}(T)}$, $\lfloor f \rfloor \implies \lceil T \rceil, \{\forall f, T \in \mathbb{R}^+ \mid f \neq 0, T \neq 0\}$. Using this, we find the smallest period to be,

$$(FFFF_{16} - FFFF_{16} + 1) \cdot 1.085 \mu\text{s} = 1.085 \mu\text{s} \quad (5)$$

From this, we use $f = \frac{1}{T} = \frac{1}{1.085} = 0.92 \mu\text{Hz}$. We do a similar process for lowest frequency.

```

        ORG 0H
        CLR P1.3

LOOP:   MOV TMOD, #01D
        MOV TH0, #0D    ; lowest frequency => highest period
        MOV TL0, #0D    ; highest period => Smallest TH, TL
        SETB P1.3
        ACALL DELAY

        CLR P1.3
        AJMP DONE

DELAY:  SETB TR0
LOOP2:  JNB TF0, LOOP2
        CLR TR0
        CLR TF0
        RET

DONE:   NOP
        END

```

Problem #5

Part A

$$\begin{aligned}
 2C_{16} + 11_{16} &\implies 2C11_{16} & (6) \\
 FFFF_{16} - 2C11_{16} + 1_{16} &= D3EF_{16} & (7) \\
 &= 54254_{10} & (8) \\
 &\implies 58\,865.59\,\mu\text{s} & (9) \\
 &= 58.87\,\text{ms} & (10)
 \end{aligned}$$

Part B

$$\begin{aligned}
 10010011_2 &\implies 93_{16} \implies 147_{10} & (11) \\
 (65536 - 147) \times 1.085 &= 70\,947.07\,\mu\text{s} & (12) \\
 &= 70.95\,\text{ms} & (13)
 \end{aligned}$$